

WEEK – 1

1. Digital Transformation Success Story of Spotify.

- Spotify is a **Swedish-origin** subscription-based music and podcast streaming service owned and operated by Spotify Technology. The platform distributes millions of songs, podcasts, and audio content from artists across all genres worldwide.
- It is available internationally in multiple languages and regions. Spotify was officially launched on **October 7, 2008** as a music streaming platform. To enjoy Spotify's services, users need to choose a Free or Premium plan.
- Its headquarters are situated in Stockholm, Sweden. Registration is required to access the service, and Spotify had over 600 million monthly active users as of early 2024.
- Spotify was founded by **Daniel Ek** and **Martin Lorentzon**. The current CEO of Spotify is Daniel Ek. Spotify began as a solution to illegal music piracy and has since transformed into the world's largest music streaming platform.
- By leveraging digital technology and data analytics, Spotify personalized its music recommendations through **AI-powered features like Discover Weekly and Daily Mixes**.
- This digital approach also helped Spotify expand into podcasts and audiobooks based on listener preferences, leading to exponential growth and dominance in the audio streaming industry.

Digital Transformation:

Digital transformation is a Strategic approach that aims to improve operational efficiency, enhance decision-making, and create new business opportunities by integrating traditionally separate IT and OT systems.

Eg: Automating attendance management system etc.

Information Technology:

IT refers to the use of computers, networks, and other physical devices, infrastructures and processes to create, process, store, secure and exchange all forms of electronic data. The commercial use of IT encompasses both computer technology and telecommunications.

Operational Technology:

It is the practice of using hardware and software to control industrial equipment and it primarily interacts with the physical world.

Eg: Manufacturing, transportation and more.

IT/OT Convergence:

- It is the integration of **information technology** systems with **Operational technology** systems.
- IT systems are used for data-centric computing while OT systems monitor events, processes, devices and make adjustments in enterprise and industrial operations.
- IT/OT Convergence involves transferring information across both **IT and OT** networks.

- This connectivity can be used to enhance the value of the systems to deliver and progress our organization further towards our business goals.

2. Case Study of Digital Transformation through IT and OT Convergence.

Amazon

Amazon, the world's largest e-commerce and cloud computing company, decided to undergo a digital transformation by integrating its Information Technology (IT) systems and Operational Technology (OT) systems across its global fulfilment centres to improve:

- Productivity
- Safety &
- Overall efficiency

Digital transformation challenges faced by Amazon:

- **Continuous evolution of customer needs** – Customers constantly demand faster delivery (same-day, next-day), better personalization, and seamless returns. Even after massive investment in digital systems, shifting customer expectations require ongoing adaptation.
- **Budget constraints** – Deploying robotic warehouses, AI logistics systems, and smart devices across hundreds of fulfilment centres worldwide requires enormous capital. Careful financial planning is essential to balance innovation costs against returns.
- **Lack of skilled manpower** – Operating advanced robotics and AI-driven supply chains requires specialized skills. Finding and training engineers, data scientists, and technicians to manage these complex systems is a significant challenge.
- **Security concerns** – As Amazon processes billions of customer transactions and integrates cloud platforms with warehouse hardware, cybersecurity risks increase. Protecting customer data, payment systems, and OT networks from breaches is critical.
- **Inflation and supply chain issues** – Rising costs of raw materials, shipping, and third-party logistics increase operational expenses. During inflationary periods, maintaining affordable pricing while managing complex global supply chains becomes increasingly difficult.

Digital transformation implementation to overcome the above difficulties:

- **Meeting customer needs** – Amazon continuously updates its recommendation engine, deploys same-day delivery drones (Prime Air), and personalises the shopping experience using machine learning to keep pace with evolving expectations.
- **Digital transformation budget planning** – Amazon creates detailed financial roadmaps for each technology deployment—robotics, cloud, AI—balancing investment against measurable returns such as reduced delivery time and lower operational costs.

- **Creating job opportunities** – Amazon retrains warehouse staff through its Upskilling 2025 programme, enabling employees to transition from manual picking roles to operating and maintaining robotic systems, addressing both manpower gaps and upskilling needs.
- **Digital transformation security** – Amazon implements end-to-end security controls across AWS infrastructure and warehouse OT networks, regularly audits systems, and trains employees on cybersecurity best practices.
- **Changing the way decisions are made** – Data and analytics are embedded in every layer of Amazon's operations. Real-time data from warehouse sensors, delivery routes, and customer behaviour drives inventory decisions, pricing strategies, and logistics optimisation.
- **Embracing Robotics and AI technologies** – Amazon deploys Kiva robots in fulfilment centres to automate shelf movement, uses AI-powered cameras for quality control, and experiments with drone delivery and autonomous vehicles to further streamline operations.

Final Outcomes

- **Increased operational visibility** – Real-time dashboards give managers instant insight into warehouse productivity and enable rapid response to operational issues.
- **Enhanced productivity** – Digital integration and robotics streamlined workflows and minimised downtime, leading to a 20% increase in overall fulfilment productivity.
- **Improved quality control** – AI-powered cameras detect damaged goods and incorrect shipments, significantly reducing defect rates and improving customer satisfaction.
- **Predictive maintenance** – Smart sensors on warehouse robots predict mechanical failures before they occur, reducing unplanned downtime by 25%.
- **Skill development** – Amazon's Upskilling 2025 initiative provides training for over 300,000 employees to adopt new technologies and systems confidently.

Conclusion

Amazon's digital transformation serves as a powerful example of how a technology-driven company can continuously reinvent its operations. By combining IT and OT convergence with robotics, AI, and data analytics, Amazon has set the benchmark for fulfilment efficiency and customer experience globally. The key takeaway for students is that digital transformation is not a one-time project but a continuous journey of innovation, with customer satisfaction and operational agility at its core.

3. Identify the Typical Process of Workflow that can be Automated.

Zomato

Zomato is an Indian multinational food delivery and restaurant aggregator company headquartered in Gurugram, Haryana. It connects customers with restaurants, enabling online food ordering and delivery. Zomato was founded in 2008 by **Deepinder Goyal and Pankaj Chaddah**. It operates across 500+ cities in India and in several international markets.

Here are some typical processes that could be automated within the context of a food-tech platform like Zomato:

1. Customer Onboarding and Registration:

- **Automated Sign-up:** AI-driven systems verify mobile numbers through OTP and auto-fill addresses using GPS, making registration seamless.
- **Profile Personalisation:** Machine learning algorithms analyse new user preferences (cuisine, diet type) from initial selections to personalise the home feed immediately.

2. Restaurant Listing and Menu Management:

- **Automated Menu Updates:** Restaurant partners can bulk-upload or sync their menus via APIs, and AI tools flag pricing inconsistencies or missing items.
- **Image Quality Check:** AI systems automatically review uploaded food images for quality and reject blurry or irrelevant images.

3. Order Processing:

- **Order Routing:** Automated systems instantly assign incoming orders to the nearest available delivery partner using real-time GPS and load-balancing algorithms.
- **Estimated Delivery Time (EDT):** AI models calculate accurate delivery estimates based on kitchen preparation time, traffic data, and delivery distance.

4. Quality Assurance:

- **Automated Review Moderation:** Natural Language Processing (NLP) tools scan user reviews for spam, offensive content, or fake reviews, removing violations automatically.
- **Restaurant Rating Algorithms:** Systems automatically update restaurant ratings based on aggregated delivery time, food quality scores, and customer feedback.

5. Marketing and Promotions:

- **Push Notification Automation:** Tools schedule personalised offers (lunchtime deals, weather-based combos) and send them at optimal times based on user behaviour.
- **Data Analytics:** Automated analytics platforms track conversion rates of promotions, helping Zomato optimise discount strategies and ad spend.

6. Delivery Partner Support:

- **Delivery Partner Chatbots:** AI-powered bots handle delivery partner queries about earnings, route issues, and payment status without human intervention.
- **Live Tracking Automation:** GPS-based systems automatically update customers on real-time delivery partner location and estimated arrival.

7. Customer Feedback and Updates:

- **Sentiment Analysis:** Automated sentiment analysis tools process user reviews to identify patterns in complaints (e.g., late delivery, cold food) and flag them for action.

- **Feature Updates:** Automated CI/CD pipelines facilitate smooth rollout of new app features and fixes without service disruption.

8. Monetisation and Analytics:

- **Subscription Management:** Zomato Gold/Pro membership billing, renewal reminders, and benefit activation are handled fully automatically.
- **Revenue Analytics:** Automated systems track restaurant commission rates, delivery charges, and advertising revenue, providing insights to improve monetisation strategies.

9. Security and Fraud Detection:

- **Automated Fraud Checks:** AI tools scan transaction patterns to detect fake orders, coupon misuse, or suspicious account activity in real time.
- **Payment Security:** Automated PCI-DSS compliance checks ensure all payment gateways meet security standards continuously.

10. Data Backup and Recovery:

- **Automated Backups:** Cloud-based automated backups of order history, restaurant data, and user profiles ensure business continuity and disaster recovery.

4. How to Create Project Plan and Product Backlog for Project and User Story Creation.

Step 1: Open the web browser and search for **Jira Login**.

Step 2: You can continue with your Gmail account or login to the Jira Tool.

Step 3: Click on **Jira Software** → **Project** → **Create Project** from the drop-down menu.

Step 4: Select **Scrum** and click on **Use Template**, then click on **Create**.

Step 5: Give a name to your project and add a Description if required.

Step 6: In the **Timeline**, create the Epic name of the project with the defined user stories.

Step 7: Now go to **Backlog**, drag and drop the User Stories and tasks, then divide them into different Sprints.

The screenshot shows the Jira Software interface for the 'IKEA App' project. The left sidebar contains a navigation menu with sections: PLANNING (IA board, Timeline, Backlog, Active sprints, Reports), Issues, Components, and DEVELOPMENT (Code). The main area is titled 'Timeline' and displays a calendar view for August and September. A list of user stories is shown on the left, including 'IA-1 IKEA', 'IA-2 Spalsh screen', 'IA-3 Displays the logo of the...', 'IA-4 Login or Sign up', 'IA-5 Login with using your g...', 'IA-6 Homepage', 'IA-7 Displays all the product...', 'IA-8 Search', and 'IA-9 Custom... searches the...'. The timeline view shows a vertical line indicating the current date. The top navigation bar includes 'Jira Software', 'Your work', 'Projects', 'Filters', 'Dashboards', 'Teams', 'Apps', and a 'Create' button. The user profile 'bharath' is visible in the top right corner.

This screenshot shows the Jira Software interface for the 'IKEA App' project, similar to the first one but with a different set of user stories. The left sidebar navigation menu is the same. The main area is titled 'Timeline' and displays a calendar view for August and September. The list of user stories includes 'IA-1 IKEA', 'IA-2 Spalsh screen', 'IA-3 Displays the logo', 'IA-10 Customers profile tab', 'IA-11 Displays the customer...', 'IA-13 Location tracking', 'IA-13 Location tracking', 'IA-13 Customers should trac...', and 'IA-15 Provides online payme...'. The timeline view shows a vertical line indicating the current date. The top navigation bar is the same, and the user profile 'bharath' is visible in the top right corner.

Step 8: Give the user story points to each user story and click on “Start Sprint”.

Step 9: Go to Active Sprint & place the User Stories and tasks to the Done State and Click on ‘Start Sprint’.

Step 10: Click on the Burn Down Chart.

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Jira Software Your work ▾ Projects ▾ Filters ▾ Dashboards ▾ Teams ▾ Apps ▾ Create 🔍 Search 🔔 ? ⚙️ 👤

IKEA App
Software project

PLANNING
IA board Board ▾
Timeline
Backlog
Active sprints
Reports
Issues
Components

DEVELOPMENT
Code
Releases
Project pages

You're in a company-managed project
[Learn more](#)

Projects / IKEA App / IA board

Search backlog 🔍 BK 👤 Only My Issues Recently Updated

Share ...

Insights

Start sprint ...

IA Sprint 2 4 issues
14/Sep/23 9:36 PM • 28/Sep/23 9:36 PM

VERSIONS

EPICS

Homepage	IKEA	IA-6	-
Displays all the products details	IKEA	IA-7	-
Search tab	IKEA	IA-8	-
Customers searches the required products	IKEA	IA-9	-

+ Create issue

4 issues Estimate 0

Backlog 6 issues

Create sprint ...

Customers profile tab	IKEA	IA-10	-
Displays all the products personal details.	IKEA	IA-11	-
Location tracking	IKEA	IA-12	-

